Wonsub Cho

Department of Physics, Sungkyunkwan University
Suwon, Republic of Korea
wikicho1674@gmail.com · wikicho.github.io · INSPIRE · github.com/wikicho

Research Interests

- Dark matter in cosmology and particle physics
- Cosmological phase transitions, gravitational waves, and baryogenesis

Education

Ph.D. Candidate in Physics Mar 2023–Feb 2027 (expected)
Sungkyunkwan University (SKKU)
Theoretical Astroparticle Physics and Cosmology Group
Advisor: Prof. Ki-Young Choi

M.S. in Physics Aug 2020–Aug 2022
Sungkyunkwan University (SKKU)
Thesis: *Effect of Boosted Dark Matter by Cosmic Particles*

B.S. in Physics Mar 2014–Feb 2020
Sungkyunkwan University (SKKU)
Thesis: *Searching for Boosted Dark Matter by High-Energy Neutrinos*

Visiting Research Experience

Visiting Student, CERN Theory Department May 2026–Present (expected end: Nov 2026)
Advisor: Jorinde van de Vis
Research: Deviations from kinetic equilibrium in electroweak baryogenesis.
Selected through the CERN–Korea Collaboration (CKC) graduate student programme.

Awards and Fellowships

2026	**CERN–Korea Collaboration (CKC) Graduate Student Programme**, selected for a research visit to CERN with support for living expenses and airfare.
2025	**Encouragement Award for Graduate Research Presentation**, SKKU Physics Workshop.
2020, 2023, 2025	**Outstanding Presentation Award**, Korean Physical Society.
2020–2024	**Graduate Merit Scholarship**, Department of Physics, SKKU.
2021–2022	**BK21 Innovation Research Fellowship**.
2019	**Excellence Award for Undergraduate Poster Presentation**, Korean Physical Society.
2018	**National Science and Technology Scholarship**, Korea Student Aid Foundation.

Teaching Experience

2025–2026	**Quantum Field Theory I and II**, Teaching Assistant, SKKU.
Fall 2024	**Electromagnetism**, Teaching Assistant, SKKU.
Fall 2023	**Relativity**, Teaching Assistant, SKKU.
2023	**Particle Physics Summer Camp**, Teaching Assistant, KIAS.
2020, 2023	**General Physics**, Teaching Assistant, SKKU.
2020–2022	**Mathematical Physics**, Teaching Assistant, SKKU.

Skills and Languages

Scientific computing Python, Mathematica
Languages Korean and English

Talks and Posters

Nov 2026 **PT@Geneva** · Talk (scheduled)
"Nanohertz gravitational waves and asymmetric dark matter from a supercooled dark sector phase transition"

Sep 2026 **DESY Theory Workshop 2026** · Talk
"Nanohertz gravitational waves and dark matter from a classically conformal dark sector"

Dec 2025 **IBS Dark Matter Axion Group Seminar** · Talk
"Reconciling cosmological tensions with inelastic dark matter and radiation in a $\mathrm{U}(1)_D$ framework"

Nov 2025 **SKKU Physics Workshop** · Poster
"Reconciling cosmological tensions with inelastic dark matter and dark radiation in a $\mathrm{U}(1)_D$ framework"

Aug 2025 **29th International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology (SI 2025)** · Poster
"Reconciling cosmological tensions with inelastic dark matter and dark radiation"

Apr 2025 **7th CUBES Workshop** · Talk
"Reconciling cosmological tensions with inelastic dark matter and dark radiation in a $\mathrm{U}(1)_D$ framework"

Apr 2025 **Spring Meeting of the Korean Physical Society** · Talk
"Reconciling cosmological tensions with inelastic dark matter and dark radiation"

Jul 2024 **IBS CTPU-CGA 2024 Summer School and Workshop for Particle Physics and Cosmology in Korea** · Talk
"Freeze-in inelastic Dirac dark matter and H_0 tension"

Apr 2024 **6th CUBES Workshop** · Talk
"Freeze-in inelastic dark matter, H_0 tension and detection prospects"

Apr 2024 **Spring Meeting of the Korean Physical Society** · Talk
"Inelastic dark matter, the H_0 tension, and detection prospects"

Jul 2023 **IBS CTPU-CGA 2023 Summer Workshop and School on Particle Physics in Korea** · Talk
"Cored dark matter halos in the cosmic neutrino background"

Jun 2023 **XVI International Conference on Interconnections between Particle Physics and Cosmology (PPC 2023)** · Talk
"Cored dark matter halos in the cosmic neutrino background"

Apr 2023 **5th CUBES Workshop** · Talk
"Cored dark halo in the cosmic neutrino background"

Apr 2023 **Spring Meeting of the Korean Physical Society** · Talk
"Cored dark matter halos in the cosmic neutrino background"

Nov 2021 **The Origin and Evolution of the Universe (APCTP)** · Talk
"Sterile neutrino dark matter with dipole interaction"

Oct 2021 **Fall Meeting of the Korean Physical Society** · Talk
"Sterile neutrino dark matter with dipole interaction"

Nov 2020 **Fall Meeting of the Korean Physical Society** · Talk
"Searching for boosted dark matter mediated by a new gauge boson"

Oct 2019 **Fall Meeting of the Korean Physical Society** · Poster
"Searching for boosted dark matter by cosmic rays"

Publications

1. **Stable dark matter from Pauli blocking in the degenerate fermion background with quantum field theory**
   Wonsub Cho, Ki-Young Choi, Junghoon Joh, and Osamu Seto
   Physics of the Dark Universe 49 (2025) 101978 · arXiv:2407.08229

2. **Reconciling cosmological tensions with inelastic dark matter and dark radiation in a $\mathrm{U}(1)_D$ framework**
   Wonsub Cho, Ki-Young Choi, and Satyabrata Mahapatra
   Journal of Cosmology and Astroparticle Physics 09 (2024) 065 · arXiv:2408.03004

3. **Cored dark matter halos in the cosmic neutrino background**
   Wonsub Cho, Ki-Young Choi, and Hee Jung Kim
   Journal of Cosmology and Astroparticle Physics 07 (2023) 013 · arXiv:2204.01431

4. **Sterile neutrino dark matter with dipole interaction**
   Wonsub Cho, Ki-Young Choi, and Osamu Seto
   Physical Review D 105 (2022) 015016 · arXiv:2108.07569

5. **Searching for boosted dark matter mediated by a new gauge boson**
   Wonsub Cho, Ki-Young Choi, and Seong Moon Yoo
   Physical Review D 102 (2020) 095010 · arXiv:2007.04555